Dhruvil Satasiya

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Objective

Machine Learning Engineer specializing in mechanistic interpretability and adversarial robustness. Proven track record of translating theoretical alignment research into optimized PyTorch infrastructure. Graduated (B.Tech) and available for immediate, full-time remote deployment.

Education

•P P Savani University, Surat *2022–May 2026 (Graduated)*

B.Tech in Information Technology
Relevant Coursework: Artificial Intelligence, Optimization Techniques, Big Data Analytics

•Indian Institute of Technology Madras (IITM), Chennai *2023– Apr 2027*

BS in Data Science
Relevant Coursework: Machine Learning Practice, Statistics, Data Structures, DBMS, Python Programming

Note: This is a flexible, asynchronous distance-learning degree that requires zero daytime hours and does not impact full-time employment availability.

Publications

Dhruvil S., F. Sojitra, R. Chauhan. "Through the Bottleneck: How Multi-head Latent Attention Separates Content from Position in Language Models." *arXiv preprint arXiv:2607.23054*, 2026. [arXiv Link] | [GitHub Repo]

Internship Experience

•AI Research Intern *June 2026-Present*

Remote AI Institute of South Carolina

- SLM Pretraining Pipeline: Architecting the end-to-end pretraining and alignment infrastructure for a novel Small Language Model (SLM).
- Data Infrastructure: Engineered the initial large-scale dataset tokenization and processing pipeline to prepare the raw corpus for the distributed pretraining run.

•Independent ML Researcher *Dec 2025-Present*

- Mechanistic Interpretability: Conducted the first comprehensive mechanistic interpretability study on Multi-head Latent Attention (MLA), utilizing SVD and linear probing on a 114M-parameter model to demonstrate how the shared c_{KV} bottleneck preserves content, discards position, and fundamentally reshapes circuit topology.
- Adversarial Robustness: Developed open-source, from-scratch PyTorch implementations of advanced white-box (Greedy Coordinate Gradient) and black-box (Prompt Automatic Iterative Refinement) red-teaming algorithms.
- Latent Architecture Research: Spearheaded a pilot study evaluating the mechanistic topography of adversarial vulnerabilities within Multi-Head Latent Attention (MLA) bottlenecks, utilizing a custom-trained 114M parameter model (drafted for ICML '26 MI workshop).

•AI/ML Intern *June 2025-Nov 2025*

On-site 10Turtle

- Designed and deployed production LLM systems with improved latency, cost efficiency, and response quality.
- Ran controlled experiments across model configurations to optimize performance in real-world settings.
- Built containerized inference services with automated logging, health checks, and monitoring to ensure >99% uptime.
- Trained and deployed YOLO-based object detection models for multi-class image analytics, optimizing data augmentation and inference pipelines.
- Converted experimental LLM workflows into production-ready systems using reproducible evaluation and logging pipelines.

•SDE Intern *Jan 2024-May 2025*

Remote Ashirwad Infotech

- Deployed a fine-tuned LLM and an open-source image generation model on GPU instances using vLLM, exposing inference APIs for low-latency serving.
- Optimized inference by resolving GPU OOM issues and evaluating 4-bit/8-bit quantization and dynamic batching configurations to improve throughput and memory efficiency.
- Fine-tuned an instruction-following LLM using SFT with LoRA/QLoRA for parameter-efficient adaptation.
- Prepared chat-style instruction datasets, including formatting, tokenization, and prompt template construction using Hugging Face and PyTorch.
- Managed the complete fine-tuning workflow from data preparation to evaluation and adapter merging.

Additional Projects

•**JEPA for Text (LeCun-inspired)**

Source Code

- Implemented predictive representation learning system to study latent structure in language.
- Investigated embedding geometry and model stability without reconstruction objectives. **Tech:** Python, PyTorch, Transformers, Scikit-learn, Matplotlib.

Technical Skills

- **Programming:** Python
- **Libraries/Frameworks:** PyTorch, Transformers, Accelerate, PEFT, NumPy, Pandas, Matplotlib, Seaborn
- **NLP & LLM Tools:** vLLM, Weights & Biases, LlamaIndex, LangChain, OpenAI APIs
- **ML Techniques:** Model Fine-tuning, A-B Testing, Object Detection, Sequence Modeling, Experiment Design
- **Deployment:** Docker, Logging/Monitoring, API Integration, Git
- **Soft Skills:** Clear technical communication, research mindset, rapid prototyping, teamwork

Certifications

•**Reinforcement Learning** NPTEL (IIT Madras) | Elite Certification *Jan-Apr 2025*

•**AWS Academy Cloud Foundations** AWS Academy *Mar 2025*